■ LASER MACHINE STARTUP PROCEDURE

HAN'S LASER (Machine 2) · Intermach Laser (Machine 1)

Always follow this procedure in order. Ensure all safety checks are completed before starting any machine.

MACHINE 2 — HAN'S LASER (FIBER LASER)

1 Turn ON the Main Rotary Switch on the Machine Body

Locate the main rotary isolator switch on the side of the HAN'S LASER machine body (CS01SHF Exator). Wearing appropriate gloves, turn the switch to the ON position to restore main power to the machine.

■ Always wear protective gloves when operating the main isolator switch.

2 Turn ON the Bypass Switch

Open the electrical cabinet door on the machine. Locate the rotary bypass switch inside the cabinet. Wearing gloves, turn the bypass switch to the ON position.

■ The cabinet contains live electrical components. Ensure gloves are worn at all times inside the cabinet.

3 Confirm the 250A Power Switch is ON

Inside the cabinet, locate the 250A Power Switch (circuit breaker labeled "Power switch"). Verify it is in the ON position. If it has tripped, reset it before proceeding. Close and secure the cabinet door.

■ If the circuit breaker has tripped, do not reset without investigating the cause first.

4 Close the Electrical Cabinet Door

Once the bypass switch and power switch are confirmed ON, close and latch the electrical cabinet door securely before proceeding to the control panel.

■ Ensure the cabinet door is fully closed and latched before the next step.

5 Turn ON the Voltage Stabiliser (SBW-150 KVA)

Locate the HAN'S LASER voltage stabiliser cabinet (SBW-150 KVA-D) positioned beside the machine. Press the power-on button on the front panel to energise the stabiliser. The KINGBO digital display will illuminate and show the regulated output voltage. Wait for the reading to settle to a stable value before proceeding to the control panel.

■ Front panel: KINGBO digital display with power-on, run and shutdown buttons. Ensure the display shows a stable output voltage before continuing.

6 Release the Emergency Stop Button

At the Machine 2 (HAN'S LASER) control panel, check the red Emergency Stop button. If it is pressed in, twist it counter-clockwise until it pops back out and releases. The control panel cannot be powered on while the Emergency Stop is engaged.

■ Ensure the work area is clear before releasing the Emergency Stop. Only release it once you are ready to power on the machine.

7 Press the POWER ON Button at the Control Panel

At the Machine 2 (HAN'S LASER) control panel, press the green POWER ON button. The Power On indicator will illuminate. The Philips CCTV monitor on top will also power on showing the laser cutting area.

■ Control panel buttons: Power On (green), Power Off (red), Emergency Stop (red mushroom), NC Start, NC Stop, Feed Override dial.

8 Wait for Windows to Boot & Log In

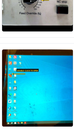
The main touchscreen will show the Windows boot sequence. Wait for the Windows desktop to fully load. If prompted, log in with the user credentials. Do not open the laser software until Windows has fully loaded.

■ The desktop will be visible once Windows has fully loaded.

9 Set the Feed Override Dial to 100%

On the control panel, locate the Feed Override (%) dial. Rotate it clockwise until the pointer reaches the 100 mark. This sets the feed rate to 100% of the programmed value.

■ Feed Override scales the programmed feed rate. Set it to 100% for normal operation; lower values slow the machine down.



10 Open the HAN'S LASER Cutting Software (LoadCutPar)

Once the Windows desktop has loaded, locate the LoadCutPar icon (the application with the two red boxes) on the desktop. Double-click it to launch the HAN'S LASER cutting software.

■ LoadCutPar is the HAN'S LASER cutting application. If it does not open on the first double-click, wait a moment and try again.

11 Wait for HAN'S LASER Software to Load

The HAN'S LASER startup screen will appear: "Starting a New World of Laser Cutting". Wait for the loading bar to complete.

■ Do not interrupt the software loading process. This may take 1–2 minutes.

12 Log in to the System

A "Log In system" dialog will appear with the User Name (USER) and Password fields. Click the checkmark (✓) button to confirm and log in to the cutting software.

■ The User Name defaults to USER. Click the checkmark to proceed; use the back arrow only if you need to cancel.

13 Activate Chiller Start, Lights & Focusing Alarm

On the HAN'S LASER main interface, find the "IH Options Activation" panel on the right-hand side. Turn on the Chiller Start, Lights and Focusing Alarm toggles so that each switch shows green (enabled).

■ Each toggle shows green when active. The chiller must be ON before cutting — if the "Chiller is not ready" warning shows at the top, wait for it to clear.

14 Machine 2 is Ready — Perform Homing if Required

The HAN'S LASER main interface will load showing the machine diagram and axis positions. If prompted with a homing message (e.g., "Cannot start until W axis homing is complete"), press the Home button via Axis Operation to home the axes before starting any job.

■ Use Axis Operation to home the machine. Homing must be completed before a cutting job can start.

15 Press HV and Fume on the Control Panel

On the Machine 2 control panel, press the HV button to enable the high-voltage supply, then press the Fume button to start the fume extraction. Both functions should be active before cutting.

■ HV enables the high-voltage supply; Fume runs the fume extraction. Confirm both are switched on. Machine 2 is now ready — proceed to start Machine 1.

MACHINE 1 — INTERMACH LASER

16 Turn ON the Main Isolator Switch (Intermach)

Move to Machine 1 (Intermach Laser). Locate the main isolator switch on the cabinet. Rotate the knob clockwise from O-OFF to the I-ON position to restore power to the machine.

■ If the switch is in the TRIPPED position, turn it to RESET (O-OFF) first, then to I-ON. Do not force the switch.

17 Press the POWER ON Button at the Intermach Panel

At the control panel, press the green POWER ON button (square). The green POWER ON indicator light will illuminate. The screen will begin to power on.

■ Intermach buttons: POWER ON (green square), POWER OFF (red square), START (green round), STOP (red round), RESET (yellow), E-STOP (large red).

18 Release the E-STOP Button (Intermach)

On the Intermach control panel, check the red E-STOP button. If it is pressed in, twist it clockwise (in the direction of the arrows) until it pops back out and releases. The machine cannot run while the E-STOP is engaged.

■ Ensure the work area is clear before releasing the E-STOP. Press RESET afterwards if the panel prompts for it.

19 Wait for Windows to Boot on Machine 1

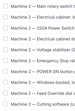
The Intermach touchscreen will display the Windows boot sequence. Wait for the Windows desktop to fully appear before opening any software. Do not press any buttons during this process.

■ The Windows desktop will appear once the PC has fully booted.

20 Open the Intermach Cutting Software (pdata)

On the Windows desktop, locate the pdata icon (the WinCC RA application at the bottom of the icon column). Double-click it to launch the Intermach laser cutting software.

■ pdata is the Intermach cutting application. If it does not open on the first double-click, wait a moment and try again.



21 Laser Cutting Software Loads — Machine 1 Ready

The Intermach software will open on the "Welcome" screen showing the 3D machine view. Tap anywhere on the screen to continue into the cutting software. The machine is now ready for operation.

■ Both machines are now ON and ready. Always perform a system check before starting a cutting job.

■ STARTUP CHECKLIST

- ☐ Machine 2 — Main rotary switch turned ON
- ☐ Machine 2 — Electrical cabinet: bypass switch ON
- ☐ Machine 2 — 250A Power Switch (circuit breaker) ON, cabinet closed
- ☐ Machine 2 — Electrical cabinet door closed and latched
- ☐ Machine 2 — Voltage stabiliser (SBW-150 KVA) ON, voltage stable
- ☐ Machine 2 — Emergency Stop released (turned counter-clockwise)
- ☐ Machine 2 — POWER ON button pressed (indicator lit)
- ☐ Machine 2 — Windows booted, logged in
- ☐ Machine 2 — Feed Override dial set to 100%
- ☐ Machine 2 — Cutting software (LoadCutPar) opened
- ☐ Machine 2 — Logged in to the system (checkmark)
- ☐ Machine 2 — HAN'S LASER software loaded
- ☐ Machine 2 — Chiller Start, Lights & Focusing Alarm activated
- ☐ Machine 2 — Homing completed (if required)
- ☐ Machine 2 — HV and Fume switched on
- ☐ Machine 1 — Main isolator switch ON (I-ON)
- ☐ Machine 1 — POWER ON button pressed
- ☐ Machine 1 — E-STOP released (turned clockwise)
- ☐ Machine 1 — Windows booted
- ☐ Machine 1 — Cutting software (pdata) opened
- ☐ Machine 1 — Cutting software loaded (Manual Mode)

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